When Does Density Beat Centroids? A Comparative Study of Clustering Methods Across Data Geometries

[**DECIDE**: final title. Proposed: "When Clustering Fails: Which Data Break Which Methods, and Can You Tell Without Labels?", to match the research question]

SC4020 Project 1 — Group 28

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Abstract

Clustering is used when there are no labels to check the result against, so a clustering is usually judged by an internal score such as the silhouette. We test whether that is safe. The methods assume different things: K-means and Gaussian mixtures expect compact groups, spectral clustering expects groups joined by chains of close neighbours, and DBSCAN and HDBSCAN expect dense groups separated by empty space. We ran all six (K-means with random and K-means++ starts, GMM, spectral clustering, DBSCAN and HDBSCAN) on seven datasets chosen to break these assumptions, five of them with known labels, and chose every setting with one label-free rule. We asked three questions: does a method fail when its assumption breaks, can the failure be seen without labels, and does extending a method by one component fix it? Every method failed where its assumption broke. K-means cut two curved clusters in the wrong place (agreement with the true labels 0.48 out of 1, against 1.00 for spectral clustering), and on housing data with no empty space between groups DBSCAN put 96% of sales in one cluster. The silhouette did not reveal these failures: on four of the five labelled datasets it favoured a different method from the one that best matched the labels, and on the curved clusters it chose a DBSCAN setting that shattered them into 72 pieces (agreement 0.01). Comparing each result with the same setting on fake data caught results with no real structure, but not K-means joining earthquake belts thousands of kilometres apart into one cluster. Adding one component to a method, such as HDBSCAN's density hierarchy or GMM's covariances, fixed the failures it targets, but none helped data with no gaps. No single score can choose a clustering method: an analyst should look at the data, report how many points each method discards, and test directly for the failures the data make likely.

1 Introduction

1.1 Motivation

Clustering is used where no answer key exists. A seismologist grouping earthquake epicentres wants active fault systems; a housing analyst grouping resale transactions wants market segments. Neither has labels to check a result against.

Each clustering method carries its own idea of what a cluster is. K-means [12] looks for compact groups around a centre and assigns every point to one of them; a Gaussian mixture allows those groups to be stretched ellipses; spectral clustering [15] joins points linked by chains of close neighbours, whatever the shape. DBSCAN [8] and HDBSCAN [5] look for dense regions separated by sparse space, and leave points in the sparse space as noise. Whether or not that idea matches the data, the method returns a partition. On one year of global earthquakes, K-means returns six tidy regions, one of which joins five plate boundaries lying 2,932 to 11,449 km apart (Section 3.4.1). Nothing in the output says that this has happened. An analyst without labels usually judges the result by an internal score such as the silhouette [17], which rates how compact and well separated the clusters are, and that score is itself built on the K-means idea of a cluster.

1.2 Research question

This report asks one question: *when a clustering method's idea of a cluster does not match the data, does it fail, and can the failure be detected without labels?* We split it into three parts.

RQ1, cause. Which property of the data breaks which method? We consider cluster shape, density that varies between clusters, the number of dimensions, the absence of gaps between groups, and values recorded in steps.

RQ2, detection. Do checks that need no labels reveal those failures? We test the silhouette, the Davies–Bouldin index, the share of points labelled noise, and a fake-data check: the same method run on data whose features have been shuffled independently, which keeps each feature's values but removes any joint structure.

RQ3, remedy. Does extending a method by one component, such as a better seeding, a density hierarchy or a covariance per cluster, repair its failures, and what does it cost?

We compare six methods, K-means with random starts, K-means++ [2], a Gaussian mixture (GMM), spectral clustering, DBSCAN and HDBSCAN, on seven datasets (Table 2), and choose every setting with one label-free rule (Section 3.3). Each dataset was chosen to test one data property, most of them a property that breaks at least one method's assumption. Five have class labels, so a method's output can be scored against them with the Adjusted Rand Index (ARI) [11], and the internal scores can be checked against that answer. Two have no labels (earthquakes, Uber pickups); there we judge failures on maps, against known geography.

1.3 Findings in brief

F1. Each method failed where the data broke its assumption. K-means and GMM split the two interleaved half-moons of D1 wrongly (ARI 0.48 and 0.50) and K-means merged distant belts on D2. With one radius, DBSCAN clustered 90% of earthquakes in the dense south-west Pacific but only 20% in the sparser Americas. In 50 dimensions (D5) it barely matched the labels (ARI 0.03). On D7, where housing sales form one mass with no gaps, DBSCAN put 96% of sales in one cluster.

F2. The silhouette did not reveal these failures. On four of the five labelled datasets, the method with the highest silhouette was not the one that matched the labels best. On D1 the rule picks a DBSCAN setting that breaks the moons into 72 small clusters (ARI 0.01) because its silhouette, 0.523, is higher than that of spectral clustering, which recovers the moons exactly (0.387, ARI 1.00). On Ecoli (D4) the silhouette orders the six methods almost exactly opposite to ARI (Spearman $\rho = -0.82$). The fake-data check does better, but only partly. It detects when a method finds nothing real: on D4 both density methods score barely above the same setting on shuffled data (0.559 against 0.545 and 0.551), and on D7 HDBSCAN scores far below it (0.158 against 0.564). It does not detect the wrong structure: on D2, K-means scores well above shuffled coordinates (silhouette 0.529 against 0.390) while merging unrelated belts. One failure was found only by a check aimed at a known problem: on D7, 34 of the 37 clusters HDBSCAN returns on the recorded values are each a single recorded storey band.

F3. Adding one component to a method repaired some failures and not others. HDBSCAN's density hierarchy raised the clustered share of the Americas from 20% to 62%, at the cost of trimming the densest belts (78% against 90%); GMM's covariance per cluster raised ARI on Ecoli from 0.508 to 0.618; K-Means++ seeding cut the mean gap to the best K-means objective on Ecoli from 52.9% to 3.4%. None of them found meaningful clusters in D7, which has no gaps.

Section 2 describes the methods and Section 3 the datasets, protocol and results, grouped by data property. Section 4 turns them into guidance.

1.4 Related work

Centroid and mixture models. Lloyd's algorithm [12] minimises the within-cluster sum of squared distances by alternating assignment and centroid updates. It converges to a local optimum that depends on the starting centres. K-means++ [2] picks each starting centre with probability proportional to its squared distance from the centres already chosen, which gives an $O(\log k)$ approximation to the optimum in expectation. A Gaussian mixture fitted by EM [7] generalises K-means to ellipsoidal clusters with soft assignments. Agglomerative clustering builds a hierarchy by repeatedly merging the closest pair of clusters. Under Ward's criterion [19] the merge cost is the increase in within-cluster variance, the same objective as K-means.

Density-based methods. DBSCAN [8] defines a cluster as a maximal set of density-connected points. A point is a core point if at least `minPts` points lie within radius ε. Points reachable from no core point are noise. The method needs no k and imposes no shape, but its single global ε assumes all clusters have similar density. OPTICS [1] orders points by reachability so that clusters at several densities can be read from one plot. HDBSCAN [5] builds a hierarchy over mutual reachability distances and keeps the clusters that persist longest. This removes ε and leaves a minimum cluster size as the main parameter.

Graph-based methods. Spectral clustering [15] embeds the points using the leading eigenvectors of a normalised affinity matrix, then runs K-means in that embedding. Groups connected through the affinity graph become compact in the embedding, so non-convex clusters can be separated. A dense affinity matrix needs $O(n^2)$ memory, which limits the method to modest n.

Validation and cluster tendency. Internal indices, such as the silhouette [17], Davies–Bouldin [6] and Calinski–Harabasz [4] indices, score a partition from the data alone. External indices such as the Adjusted Rand Index [11] compare it with known labels. Von Luxburg et al. [13] argue that no criterion can judge a clustering apart from its intended use. A separate line of work tests whether data contain clusters at all. The Hopkins statistic [9] compares nearest-neighbour distances in the data with those in uniformly random points. The gap statistic [18] compares within-cluster dispersion with its value on reference data drawn from a null distribution. Our fake-data check applies the same idea to every method we compare. In high dimensions the ratio of nearest to farthest neighbour distance tends to one [3], which weakens any method built on a density threshold.

2 Methods

2.1 K-Means and K-Means++

Overview K-means [12] looks for k centres $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_k$ that minimise the within-cluster sum of squares (inertia), $J = \sum_{i=1}^{n} \min_j \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2$. Each iteration assigns every point to its nearest centre and moves each centre to the mean of its points. J never rises, so the method stops at a local optimum that depends on the starting centres. One iteration costs $O(nkd)$. Random starts pick k data points uniformly. K-means++ [2] picks the first centre at random and each next one with probability proportional to its squared distance from the nearest centre already chosen. Only the start differs, so we treat the two as one method with two initialisations (Ablation A).

Implementation We use scikit-learn's `KMeans` with 10 starts; the only parameter is k. Because J sums squared Euclidean distances, K-means suits compact, roughly round clusters of similar spread. It cuts curved or elongated clusters, gives every point a cluster, so outliers pull the centres, and depends on the scale of each feature.

2.2 DBSCAN

Overview DBSCAN [8] takes a radius ε and a count `minPts`. A point is a core point if at least `minPts` points lie within ε of it. Core points within ε of each other join the same cluster, points within ε of a core point join its cluster as border points, and every other point is noise. The method needs no k and assumes no shape, and on coordinates ε is a distance on the ground that an analyst can judge. With a spatial index a fit costs about $O(n \log n)$ in few dimensions, rising towards $O(n^2)$ in many dimensions or with a large ε. A reference value for ε comes from the k-distance curve, each point's distance to its `minPts`-th nearest neighbour in sorted order; its knee is the point farthest from the line joining the two ends of the curve. The weakness is the single radius. It assumes all clusters have similar density, so a radius that separates dense clusters calls sparse ones noise, and one that keeps sparse clusters merges dense ones. In many dimensions distances become alike [3], and a radius separates little.

2.3 HDBSCAN

Overview HDBSCAN [5] removes ε. Each point's core distance is the distance to its m-th nearest neighbour, and the mutual reachability distance $d_{\mathrm{mr}}(a, b) = \max\{\mathrm{core}(a), \mathrm{core}(b), d(a, b)\}$ pushes points in sparse regions apart. A minimum spanning tree under d_{mr} holds the DBSCAN clusterings at every density level at once. The tree is condensed by dropping any split that leaves fewer than `min_cluster_size` points, and the clusters that persist over the widest range of density are kept; the remaining points are noise. Different branches can thus be cut at different densities, which is what DBSCAN cannot do. We use scikit-learn's `HDBSCAN`; the main parameter is `min_cluster_size`, and m (`min_samples`) defaults to it. A fit costs between $O(n \log n)$ and $O(n^2)$. Because `min_cluster_size` is a count, the result changes with the size of the sample. HDBSCAN can call large parts of the data noise, and like DBSCAN it has nothing to find where the data have no density gaps.

2.4 Gaussian Mixture Model

Overview A Gaussian mixture model (GMM) treats the data as drawn from k Gaussians, $p(\mathbf{x}) = \sum_{j=1}^{k} \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$, and fits the weights π_j, means and covariances by expectation–maximisation (EM) [7]. The E-step computes each point's responsibility (posterior probability) under every component; the M-step re-estimates π_j, $\boldsymbol{\mu}_j$ and $\boldsymbol{\Sigma}_j$ from those soft assignments; the two steps repeat until the log-likelihood stops improving. Each point is then given its most probable component.

Implementation We use scikit-learn's `GaussianMixture` [16] with k-means initialisation, three starts and a fixed seed. The parameters are the number of components k, which is the same k as K-means on each dataset (Section 3.3), and the covariance type (full, tied, diagonal or spherical), chosen by the lowest BIC at that k; full covariances won on every dataset except D1 (tied) and D6 (diagonal). One EM iteration costs $O(nkd^2 + kd^3)$. GMM generalises K-Means: full covariances allow ellipsoidal clusters of different size, orientation and weight, but each cluster must still be a single convex blob. It fails on curved or nested shapes, has no noise label, and with full covariances it needs many points per component when d is large.

2.5 Spectral clustering

Overview Spectral clustering [15] turns clustering into graph partitioning. It (i) builds an affinity matrix W whose entry w_{ij} measures how similar points i and j are; (ii) forms the normalised graph Laplacian and takes its k leading eigenvectors, giving each point a k-dimensional embedding; (iii) runs K-Means on the rows of that embedding. Points joined by a chain of close neighbours land

near each other in the embedding even when they are far apart in the input space, so the method can recover curved and non-convex clusters.

Implementation We use scikit-learn's `SpectralClustering`, with a symmetric graph that links each point to its 10 nearest neighbours as the affinity on every dataset, and the same k as K-means. The graph adapts to the local scale of the data, where one global RBF bandwidth fits poorly, and it is sparse: a dense RBF affinity costs $O(n^2 d)$ time and $O(n^2)$ memory and a full eigendecomposition $O(n^3)$, which would rule out the 15,000 and 20,000 points of D3 and D7, while the sparse graph with an iterative eigensolver fits them in under 1.2 s (Section 3.6). Spectral clustering needs k, has no noise label, and depends on the neighbour count.

Table 1: Structural assumptions of the methods compared.

	K-Means	**K-Means++**	**DBSCAN**	**HDBSCAN**	**GMM**	**Spectral**
Cluster shape assumed	isotropic	isotropic	arbitrary	arbitrary	ellipsoidal	connected (graph)
Requires k	yes	yes	no	no	yes	yes
Explicit noise label	no	no	yes	yes	no	no
Handles varying density	no	no	no	yes	partly[b]	partly[c]
Parameter is interpretable	no	no	yes (km)	partly	no	no
Time complexity	$O(nkdi)$	$O(nkdi)$	$O(n \log n)$[a]	$O(n \log n)$[a]	$O(nkd^2 i)$	$O(n^2 d + n^3)$[d]

[a] With an effective spatial index; $O(n^2)$ worst case. [b] Per-component covariance and weight. [c] With a k-NN affinity, which adapts to local scale. [d] Dense RBF affinity and full eigendecomposition; lower with a sparse k-NN graph.

3 Experiments

3.1 Datasets

Table 2: Datasets, grouped by the data property each one tests.

ID	**Dataset**	n	d	**Labels**	**Property tested**
D1	make_moons (synthetic)	1,000	2	2 classes	curved shapes (control)
D2	Earthquakes, 2023, M≥4.5	7,643	2	none	long belts, uneven density
D3	Uber NYC pickups, Apr 2014	15,000	2	none	density contrast in a city
D4	Ecoli	336	7	8 classes	dimensionality, rare classes
D5	Fashion-MNIST (PCA)	3,000	50	10 classes	high dimensions
D6	Wine	178	13	3 classes	silhouette vs labels
D7	HDB resale flats, 2017–2026	20,000[b]	4	flat type (7)	no gaps, scale, recorded ties

[b] Random sample of 234,292 usable sales; all 234,292 used for runtime (Section 3.6).

D1 make_moons. Two interleaving half-circles from scikit-learn's `make_moons` [16]: 1,000 points, Gaussian noise 0.07, seed 42, then standardised. The two classes are dense, equal in size and clearly separated by a gap, but neither is convex: each moon wraps around the end of the other, so the straight boundary a centroid method draws must cut both. With the true labels known and no other difficulty present, D1 isolates cluster shape as the single property under test.

D2 Earthquakes. Every event of magnitude 4.5 or more in 2023 from the USGS ComCat catalogue [22]: 7,651 events, of which 8 volcanic eruptions near Saipan are removed, leaving 7,643 earthquakes. We cluster the epicentre only. Depth is left out because 44.2% of events carry the default depth of 10 km. Each method meets a different difficulty. Earthquakes lie along narrow belts thousands of kilometres long, which breaks the compact-cluster assumption of K-means. Density varies widely: the distance to the nearest earthquake runs from 1.2 km (5th percentile)

to 100.5 km (95th), so no single DBSCAN radius suits every belt. And 17.2% of earthquakes lie within 10° of the 180° meridian, where flat coordinates cut the map. There are no labels.

D3 Uber NYC. Uber pickups in New York City in April 2014, from the trip records the city's Taxi and Limousine Commission released to FiveThirtyEight [25]: 564,516 pickups, each a latitude and longitude. We keep the 558,019 inside a box around the city (latitude 40.57 to 40.92, longitude −74.15 to −73.70), which drops GPS errors, and sample 15,000 at random (seed 42). The coordinates are projected to kilometres about the mean latitude, so ε is a distance on the ground. Pickups crowd into lower and midtown Manhattan (71% of the sample), with smaller dense spots at the two airports and sparse demand in the outer boroughs. The distance to the 4th nearest pickup runs from 0.008 km to 0.256 km (5th to 95th percentile), but the dense and sparse areas meet with no clear gap between them. There are no labels.

D4 Ecoli. The UCI Ecoli data [10]: 336 proteins of *E. coli*, each described by seven scores computed from its amino-acid sequence and labelled with one of eight sites in the cell where the protein is found. Features are standardised. The classes are very unequal: the five largest hold 143, 77, 52, 35 and 20 proteins, the three smallest 5, 2 and 2. D4 is hard for density methods in two ways. Seven dimensions already spread the points so that a density threshold separates less well, and the rare classes are too small to form a dense region of their own, so a density method tends to call them noise.

D5 Fashion-MNIST. Fashion-MNIST [20] holds 70,000 28×28 greyscale images of 10 clothing classes. We pool the train and test files, scale pixels to $[0, 1]$, draw a class-stratified sample of 3,000 images (300 per class, seed 42), and project the 784 pixels onto 50 principal components, which keep 86.5% of the variance. The sample keeps Spectral, DBSCAN and HDBSCAN within reach on a laptop (Section 3.6) while keeping the classes balanced. D5 is hard because the space is high dimensional, so distances concentrate [3], and because several classes (shirt, T-shirt, pullover, coat) overlap heavily with no density gap between them.

D6 Wine. The UCI Wine data [24]: 178 wines grown in one Italian region, each described by 13 chemical measurements such as alcohol, malic acid, flavanoids and proline. The cultivar (3 classes of 59, 71 and 48 wines) is held out as the label. Features are standardised, because proline runs from 278 to 1,680 while hue stays below 2. The classes are compact, so K-means recovers them well (ARI 0.897). The dataset is in the study for the opposite reason. No sparse gap separates the cultivars, so the density methods return 2 clusters with a quarter to two fifths of the wines as noise, and the silhouette still prefers them (Section 3.4.5).

D7 HDB resale. Every resale of a public housing flat in Singapore from January 2017, from data.gov.sg [23]. The 8,245 most recent sales (3.4%) have no price index value yet and are dropped, leaving 234,292. Methods are compared on a random sample of 20,000 (seed 0); all 234,292 are used for runtime. There are four features: floor area, remaining lease, storey, and price adjusted to 2026 levels with the HDB resale price index. Flat type (7 classes) is held out as a label, but it is nearly a function of size: K-means on floor area alone reaches ARI 0.632 against it. The data are hard for density methods. There are no gaps between groups of flats, and storeys are recorded in 3-storey bands, so 77.7% of sales share their floor area, lease and storey with at least one other sale.

3.2 Preprocessing

Three rules apply wherever a dataset has the property they address.

Coordinates on the sphere. Latitude and longitude treated as flat numbers put two earthquakes 6.07 km apart at 359.94° of longitude from each other, because they lie on either side of the 180° meridian. On D2, density methods therefore use the haversine (great-circle) distance on coordinates in radians, while K-means, GMM and spectral clustering work on each epicentre as a 3D point on the unit sphere. The straight-line distance between two such points grows with their great-circle distance, so the clustering follows surface distance. D3 covers half a degree of one city, where a projection to kilometres is accurate.

Standardised features. Features in their raw units are not comparable. On D7, price (in dollars) accounts for 100.0% of the squared distance between sales, so the other three features are ignored. Every feature is scaled to mean 0 and standard deviation 1.

Values recorded in steps. When a feature is recorded in steps, many points share exact values, and density methods read each shared value as a dense spot. Before running a density method on D7, each stepped value is moved by a uniform random amount of up to half its step: ± 1.5 storeys, $\pm 0.5\,\mathrm{m}^2$ of floor area and $\pm\frac{1}{24}$ year of lease. Price is continuous and is left unchanged.

3.3 Evaluation metrics and parameter selection

Metrics. Every result table reports the same columns, computed by one shared scoring function. *Internal* scores need no labels. The silhouette coefficient [17] of a point is $(b-a)/\max(a,b)$, where a is its mean distance to its own cluster and b its mean distance to the nearest other cluster; we report the mean over points, from -1 to 1, higher is better. On datasets above 5,000 points (D2, D3, D7) it is estimated on a random sample of 5,000. On D2 it uses the great-circle distance. The Davies–Bouldin index [6] averages, over clusters, the worst ratio of within-cluster spread to between-centre distance; lower is better. Both are computed on clustered points only, so they say nothing about the points a density method calls noise. Two *coverage* figures therefore stand next to them: the noise fraction and the share of points in the largest cluster. *External* scores need labels. The Adjusted Rand Index (ARI) [11] counts pairs of points on which a clustering and the labels agree, corrected so that a random clustering scores about 0 and a perfect one 1. Normalised mutual information (NMI) measures the same agreement in terms of shared information. For both, noise counts as one more cluster, so a method cannot raise its score by discarding hard points. We also report the fit time.

Fake-data check. An internal score has no fixed scale: a silhouette of 0.3 can be strong on one dataset and meaningless on another. Each chosen setting is therefore run again on fake data that keeps each feature's distribution but has no joint structure: every column shuffled independently. For density methods on coordinates (D2, D3) the fake points are instead drawn uniformly in the same range, because shuffling latitude and longitude keeps the dense bands of each. A result counts as structure only if the real data score clearly above the fake data. On D1 and D6 the setting is chosen again on the fake data rather than reused, so the check asks whether the rule finds anything there at all.

Selection rule. One label-free rule picks every setting (`sc4020.tuning`).

- **DBSCAN and HDBSCAN:** among the settings that give at least 2 clusters and label at most 50% of points as noise, the one with the highest silhouette. The noise cap is needed because the silhouette ignores noise. Without it, the silhouette prefers settings that throw most of the data away: on D2 it would pick DBSCAN with $\varepsilon = 20.4$ km, which scores 0.878 by calling 69.4% of earthquakes noise; on D7 it would pick HDBSCAN with `min_cluster_size` 400, which scores 0.451 with 71.9% of sales as noise.

- **K-means:** on labelled data, the number of classes. Otherwise, among the K whose seeds agree (minimum pairwise ARI of at least 0.95 over 10 seeds), the one with the largest gap between the real silhouette and the silhouette of the same K on fake data. The silhouette alone would pick $K = 2$ on both D2 and D3, where fake data score almost as well (0.544 against 0.528 on D2).
- **GMM and spectral clustering:** the K of K-means on the same dataset, so that the methods differ only in how they cluster; GMM's covariance type by BIC.

On labelled data we also report the setting with the best ARI, so the cost of choosing without labels is visible.

Parameter grids. K-means and K-means++ run 10 starts at each K, over $K = 2$ to 12 (D2, D7), 2 to 25 (D3, D4) or 2 to 10 (D1, D5, D6). DBSCAN's radius ε comes from the k-distance curve, the sorted distance from each point to its `minPts`-th neighbour: on D2 at the 25th to 95th percentiles of the curve (11 to 810 km) with `minPts` in $\{5, 10, 20\}$; elsewhere at multiples of the curve's knee (0.5 to 2 on D7, 0.1 to 3 on D3 and D4, 0.5 to 4 on D1, D5 and D6), with `minPts` in $\{5, 10, 20\}$ on D7, $\{4, 8, 16, 32\}$ on D3, $\{3, 5, 8, 12\}$ on D4, and $m/2$, m, $2m$ with $m = \min(\max(2d, 4), 20)$ on D1, D5 and D6. HDBSCAN's `min_cluster_size` is scaled to the data (25 to 100 on D2, 100 to 400 on D7, 10 to 250 on D3, 5 to 30 on D4, 5 up to $n/10$ on D1, D5, D6), with `min_samples` in {default, 5, 10} except on D3 and D4 (default only). The datasets keep the grids their owners chose, because a sensible range depends on n, d and the units, but the rule that picks from each grid is the same.

Reproducibility. All experiments run from the notebooks in `experiments/` on the shared package `sc4020`, with Python 3.13.16, scikit-learn [16] 1.9.1 and NumPy 2.5.3, on an Intel Core Ultra 5 225 with 32 GB of memory (D5 was run by its owner on the same versions). Random steps use seed 0 on D2 and D7 and seed 42 elsewhere; repeated runs use seeds 0 to 19 (D2, D7) or 0 to 29 (D3, D4).

3.4 Comparison by data property

Tables 3–5 give every method on every dataset in the same columns. S: silhouette; DB: Davies–Bouldin, both on clustered points only; Largest: share of points in the largest cluster; Null S: silhouette of the same setting on fake data (Section 3.3); "none": fewer than 2 clusters formed on the fake data. Density settings are given as (ε, `minPts`) and (`min_cluster_size`). Bold: the best silhouette and the best ARI per dataset among the settings chosen without labels.

Table 3: D2 and D7 at the settings chosen in Section 3.3. D2 has no labels; on D7 ARI is against flat type. K-means rows use one start, as in Ablation A.

Data	Method	Clusters	Noise	Largest	S	DB	ARI	NMI	Time (s)	Null S
D2	K-Means ($K = 6$)	6	0%	44.3%	0.525	0.800	–	–	0.003	0.389
D2	K-Means++ ($K = 6$)	6	0%	31.0%	0.529	0.738	–	–	0.008	0.390
D2	GMM ($K = 6$, full)	6	0%	28.2%	0.407	1.051	–	–	0.196	0.364
D2	Spectral ($K = 6$)	6	0%	31.6%	0.333	0.914	–	–	0.096	0.320
D2	DBSCAN (101.8 km, 20)	41	40.4%	11.8%	0.703	0.351	–	–	0.044	none
D2	HDBSCAN (50)	28	30.7%	7.5%	**0.758**	0.329	–	–	1.041	0.320
D7	K-Means ($K = 7$)	7	0%	25.2%	**0.301**	1.106	**0.223**	0.341	0.009	0.191
D7	K-Means++ ($K = 7$)	7	0%	24.8%	0.257	1.258	0.173	0.309	0.014	0.191
D7	GMM ($K = 7$, full)	7	0%	22.9%	0.126	2.225	0.158	0.269	1.158	0.084
D7	Spectral ($K = 7$)	7	0%	36.5%	−0.048	2.658	0.086	0.197	0.489	−0.018
D7	DBSCAN (0.589, 20)	2	2.6%	96.0%	0.249	0.889	0.033	0.088	0.517	none
D7	HDBSCAN (200)	2	31.6%	65.6%	0.158	1.103	0.076	0.105	2.442	0.564

Table 4: D3 and D4. Top: settings chosen without labels. Bottom: D4 settings with the best ARI, which need the labels. D3 has no labels.

Data	Method	Clusters	Noise	Largest	S	DB	ARI	NMI	Time (s)	Null S
D3	K-Means ($K = 5$)	5	0%	44.5%	0.481	0.699	–	–	0.030	0.361
D3	K-Means++ ($K = 5$)	5	0%	44.5%	0.481	0.699	–	–	0.036	0.362
D3	GMM ($K = 5$, full)	5	0%	40.2%	0.454	0.866	–	–	0.121	0.384
D3	Spectral ($K = 5$)	5	0%	35.3%	0.155	1.389	–	–	0.193	0.291
D3	DBSCAN (1.53 km, 32)	2	2.1%	95.9%	**0.779**	0.171	–	–	0.482	none
D3	HDBSCAN (250)	3	4.8%	91.2%	0.641	0.266	–	–	1.271	0.343
D4	K-Means ($K = 8$)	8	0%	20.8%	0.236	1.135	0.379	0.585	0.017	0.187
D4	K-Means++ ($K = 8$)	8	0%	23.8%	0.260	1.091	0.508	0.641	0.022	0.195
D4	GMM ($K = 8$, full)	8	0%	38.4%	0.157	2.024	**0.618**	0.597	0.040	0.159
D4	Spectral ($K = 8$)	8	0%	18.5%	0.231	1.305	0.392	0.591	0.025	0.173
D4	DBSCAN (2.50, 3)	2	0.3%	97.0%	**0.559**	0.637	0.038	0.118	0.002	0.545
D4	HDBSCAN (5)	2	0.3%	97.0%	**0.559**	0.637	0.038	0.118	0.002	0.551
D4	K-Means++ ($K = 7$), best ARI	7	0%	43.5%	0.354	0.861	0.731	0.679	0.021	0.195
D4	GMM ($K = 7$), best ARI	7	0%	38.1%	0.198	1.557	0.618	0.598	0.043	0.174
D4	Spectral ($K = 7$), best ARI	7	0%	26.5%	0.241	1.261	0.445	0.591	0.024	0.174
D4	DBSCAN (0.98, 12), best ARI	2	25.0%	50.6%	0.450	0.875	0.479	0.463	0.001	0.320
D4	HDBSCAN (12), best ARI	2	37.2%	44.3%	0.483	0.789	0.339	0.376	0.003	none

Table 5: D1, D5 and D6, with K set to the number of classes. Random K-means starts give the same partition as K-Means++ on D1 and D6 and ARI 0.344 on D5. Null S was not computed for D5 (see Limitations).

Data	Method	Clusters	Noise	Largest	S	DB	ARI	NMI	Time (s)	Null S
D1	K-Means++ ($K = 2$)	2	0%	50.5%	0.495	0.812	0.481	0.383	0.018	0.353
D1	GMM ($K = 2$, tied)	2	0%	50.0%	0.495	0.812	0.501	0.400	0.012	0.353
D1	Spectral ($K = 2$)	2	0%	50.0%	0.387	1.023	**1.000**	1.000	0.024	0.319
D1	DBSCAN (0.049, 4)	72	47.1%	1.8%	0.523	0.565	0.009	0.205	0.002	0.271
D1	HDBSCAN (100)	2	15.6%	42.4%	**0.529**	0.735	0.709	0.676	0.012	0.536
D5	K-Means++ ($K = 10$)	10	0%	16.0%	**0.202**	1.470	0.343	0.512	0.335	–
D5	GMM ($K = 10$, full)	10	0%	18.3%	0.129	2.497	0.389	0.572	33.97	–
D5	Spectral ($K = 10$)	10	0%	20.0%	0.194	1.568	**0.398**	0.586	1.402	–
D5	DBSCAN (4.42, 40)	2	22.7%	75.7%	0.138	1.211	0.028	0.103	0.019	–
D5	HDBSCAN (15)	2	32.9%	66.3%	0.039	1.754	0.023	0.064	0.384	–
D6	K-Means++ ($K = 3$)	3	0%	36.5%	0.285	1.389	0.897	0.876	0.014	0.066
D6	GMM ($K = 3$, diagonal)	3	0%	39.3%	0.280	1.412	**0.915**	0.892	0.010	0.047
D6	Spectral ($K = 3$)	3	0%	36.0%	0.283	1.391	0.880	0.861	0.021	0.038
D6	DBSCAN (3.09, 40)	2	23.6%	53.4%	**0.365**	1.080	0.431	0.536	0.001	none
D6	HDBSCAN (5)	2	42.1%	42.1%	0.363	1.046	0.260	0.354	0.002	none

3.4.1 Curved and elongated clusters

On D1 the centroid methods fail and the graph method succeeds. K-Means++ (ARI 0.481) and GMM (0.501) both draw a straight boundary that cuts each moon in two, giving the tip of one moon to the other cluster (Figure 7). Spectral clustering with a 10-nearest-neighbour graph recovers both moons exactly (ARI 1.000). The density methods can do so too: their best settings by ARI reach 1.000. But the rule does not pick them. It picks a DBSCAN radius so small that the moons break into 72 fragments with 47% of points as noise (ARI 0.009), and an HDBSCAN setting that drops the sparse moon tips as noise (15.6%, ARI 0.709). The silhouette ranks the methods almost the other way round: HDBSCAN 0.529, DBSCAN 0.523, K-Means++ and GMM 0.495, and spectral clustering last at 0.387, because it rewards compact, convex groups, which is exactly the wrong assumption here.

K-means assumes compact clusters, and D2 breaks that assumption at planetary scale. Earthquakes follow plate boundaries, so a cluster is a long, narrow belt rather than a ball around a centre. K-Means++ with $K = 6$ must cover the whole globe with six roughly round regions, so each region takes in several unrelated belts (Figure 1). One cluster of 905 earthquakes joins

Turkey, the Reykjanes Ridge and Iceland, Afghanistan and Tajikistan, Somalia and Eritrea, and the Mid-Indian Ridge. HDBSCAN finds each of these as its own cluster, and their centres are 2,932 to 11,449 km apart. K-means also has no noise label, so every isolated earthquake in the oceans is assigned to some region. Its silhouette is still 0.529, well above the 0.390 of shuffled coordinates: the score confirms that the data are not random, not that the clusters are meaningful. GMM and spectral clustering, held to the same six clusters, score lower (0.407 and 0.333); with six clusters for the 28 belts HDBSCAN finds, every one of them must also join several belts.

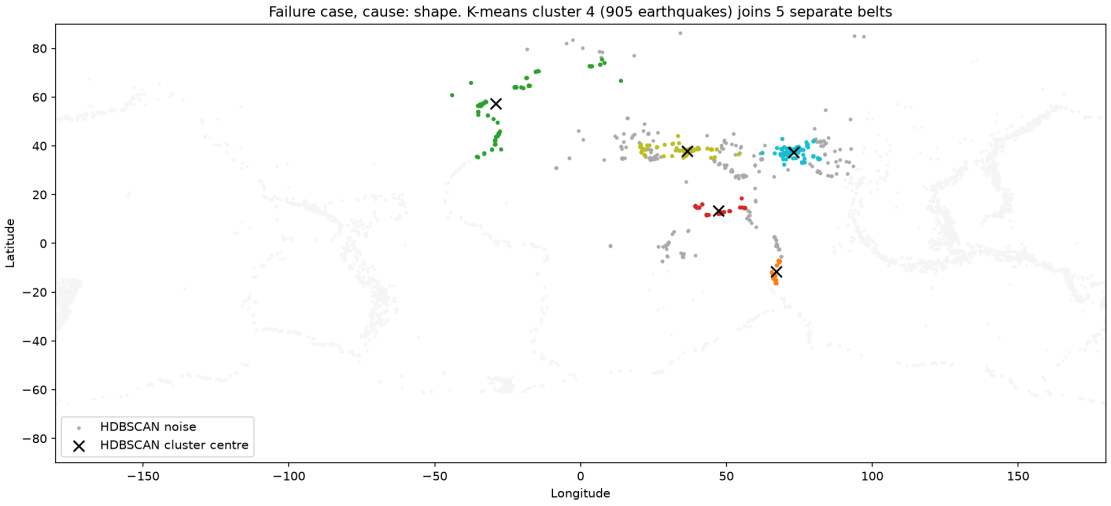


Figure 1: D2, failure of K-Means++ ($K = 6$). All points shown in colour or dark grey belong to one K-means cluster; colours are the HDBSCAN clusters inside it, crosses their centres. Cause: cluster shape. Belts thousands of kilometres apart are joined because K-means can only draw compact regions.

3.4.2 Uneven density

On D2 the distance to the nearest earthquake runs from 1.2 km to 100.5 km (5th to 95th percentile), and the k-distance curve rises smoothly with no knee. Any single ε is therefore too small for some belts and too large for others. At the chosen $\varepsilon = 101.8$ km, DBSCAN clusters 90% of the earthquakes in the dense south-west Pacific but only 20% in the sparser Americas (Figure 2). HDBSCAN, which picks a density level per cluster, clusters 62% of the Americas and keeps the Mexico to Panama belt and the Chile and Argentina belt as long clusters. The price is paid in the dense region, where it clusters 78% instead of 90%. Overall HDBSCAN keeps more earthquakes (69.3% against 59.6%) with a higher silhouette (0.758 against 0.703, Table 3).

On D3 density varies as much, but inside one connected mass. Lower and midtown Manhattan, the rest of Manhattan, Brooklyn and western Queens meet without a gap, so the density methods keep them together. DBSCAN at the knee of the k-distance curve ($\varepsilon = 0.363$ km, `minPts` 4) returns 48 clusters, yet 81.4% of pickups sit in the first and 9.6% in a second one over Brooklyn; the rest are the airports and dozens of small spots, and the silhouette is -0.07. The rule prefers a radius four times larger, $\varepsilon = 1.53$ km: two clusters, the city (95.9%) and JFK, with a silhouette of 0.779 (Table 4). HDBSCAN finds the city, JFK and LaGuardia (91.2%, 2.0% and 1.9%; 4.8% noise). The airports are the only parts of D3 with a gap around them, and both density methods find them. K-means ($K = 5$) does divide the core, into southern Manhattan (44.5%), northern Manhattan with the Bronx (39.9%) and Brooklyn (9.3%), plus the two airport areas, but those borders follow the K-means objective, not gaps in the data. Its silhouette is 0.481 against 0.362 on shuffled coordinates, so the split is not random, but nothing in the data marks where it should fall. Spectral clustering, which cuts the neighbour graph where it is thinnest, scores below its own fake data (0.155 against 0.291). The likely reason is that a mass with no thin places gives it nowhere natural to cut.

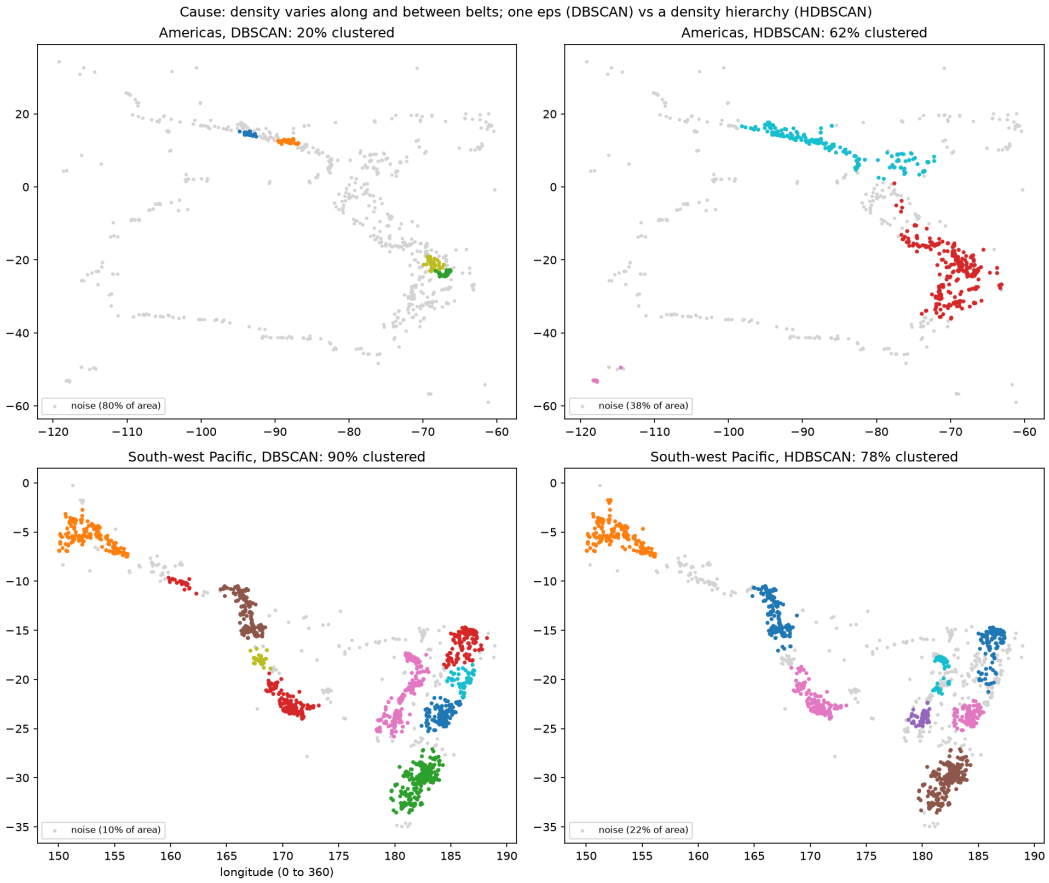


Figure 2: D2, DBSCAN ($\varepsilon = 101.8$ km) against HDBSCAN in a sparse region (top, the Americas) and a dense one (bottom, the south-west Pacific); grey points are noise. Success of HDBSCAN on the sparse belts, at the cost of trimming the dense ones. Cause: density varies along and between belts.

3.4.3 High dimensions

On D4 (seven dimensions) the methods that need K do best: GMM (ARI 0.618), K-Means++ (0.508) and spectral clustering (0.392) at the number of classes. The density methods do far worse. The rule picks settings for DBSCAN and HDBSCAN that give the same partition, one cluster holding 97% of proteins and ARI 0.038 (Table 4), and their silhouette (0.559) is barely above the same settings on shuffled columns (0.545 and 0.551). The fake-data check flags both. Even with the labels, the best DBSCAN setting reaches only 0.479, by calling a quarter of the proteins noise.

On D5 ($d = 50$) the gap is wider. Spectral (ARI 0.398), GMM (0.389) and K-Means++ (0.343) all fall well short of the labels because shirts, T-shirts, pullovers and coats overlap. DBSCAN finds two clusters and 23% noise (ARI 0.028): one cluster absorbs three quarters of the images. HDBSCAN also finds two clusters, with 33% noise (ARI 0.023). Even the best density settings by ARI reach only 0.034 and 0.066, because in 50 dimensions the k-distances of all points are similar and there is no density gap between the overlapping classes for one threshold to find.

3.4.4 No gaps between groups

D7 is one continuous mass: flats come in every combination of size, lease, storey and price, with no empty space between groups. Table 3 shows what each method does with it. K-means still finds a weak partition into seven segments, with a silhouette of 0.301 against 0.191 on shuffled columns. That ratio of 1.58 is modest, but it is the clearest result on D7 that beats its fake data; GMM's 0.126 against 0.084 is weaker, and spectral clustering scores below zero on both. Density methods have no sparse region to cut along. The DBSCAN setting chosen by the rule puts 96.0% of sales in

one cluster, and on shuffled columns the same setting forms a single cluster too. HDBSCAN finds one core holding 65.6% of sales and calls 31.6% noise, and it scores far below the same setting on shuffled columns (0.158 against 0.564). On the recorded values, without the jitter of Section 3.2, HDBSCAN instead returns 37 clusters, 34 of them single storey bands. On data with no gaps, a density method has nothing to find, and the selection rule cannot tell.

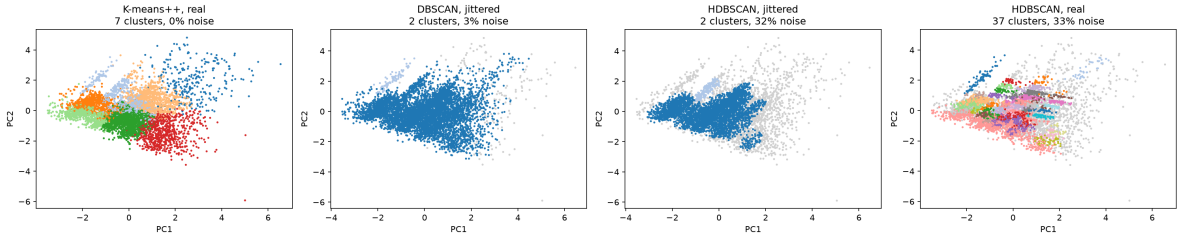


Figure 3: D7 at the chosen settings, drawn on the first two principal components (6,000 sales; grey: noise). K-Means++ cuts the mass into seven segments; DBSCAN and HDBSCAN on jittered values return one dominant cluster; HDBSCAN on recorded values (right) returns 37 clusters, most of them single storey bands. Causes: no gaps between groups, and values recorded in steps.

3.4.5 Silhouette versus ground truth

On four of the five labelled datasets the method with the highest silhouette is not the one that matches the labels best. On D1 the top silhouette goes to HDBSCAN (0.529) and the best ARI to spectral clustering (1.000, silhouette 0.387). On D4 the silhouette ranks the six methods almost exactly opposite to ARI (Spearman $\rho = -0.82$; Figure 4 shows the four main ones): the two density methods top it with the worst ARI. On D5 K-Means++ has the top silhouette (0.202) and spectral clustering the best ARI (0.398). D7 is the only exception, and a weak one: K-means with random starts has both the top silhouette (0.301) and the top ARI against flat type (0.223). In each case the silhouette favours a result that discards points or draws compact groups, not the one that follows the classes.

D6 Wine and the fake-data check. D6 repeats the pattern inside the noise cap. DBSCAN and HDBSCAN pass the cap with a quarter and two fifths of the wines as noise, take the two highest silhouettes (0.365 and 0.363) and the two lowest ARI (0.431 and 0.260), while GMM, K-Means++ and spectral clustering reach 0.915, 0.897 and 0.880 with silhouettes near 0.28. The fake-data check does not separate good from bad here either. Every method that assigns all points scores far above its shuffled-column silhouette (0.285 against 0.066 for K-Means++), and the density methods form fewer than two clusters on shuffled data. So each result passes the check, including the density results that match the labels worst: what they find, a dense core of one class, is real structure, but it is not the class structure. The check shows that a result is not an artefact of the marginals. It does not show which of several real structures a method has found.

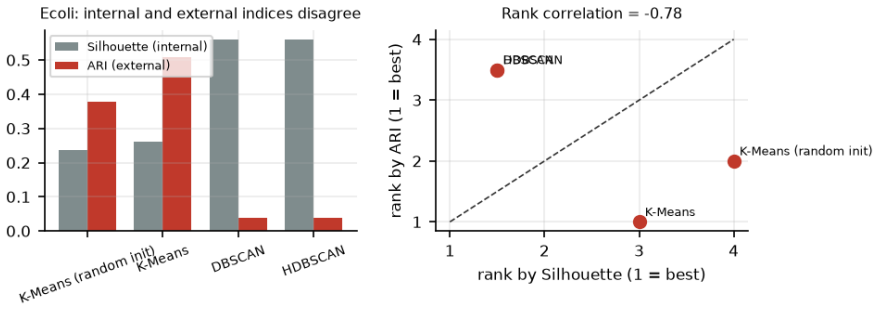


Figure 4: D4, silhouette against ARI for the four main methods at the settings chosen without labels. Left: the two scores. Right: their ranks; points on the dashed line would mean the two agree (DBSCAN and HDBSCAN coincide). The silhouette ranks the methods almost in reverse (Spearman $\rho = -0.78$; -0.82 with GMM and spectral clustering). Cause: the silhouette ignores noise and favours compact clusters.

3.5 Ablation of method components

An ablation removes or swaps one component of a method and keeps everything else fixed. Four of our methods extend a simpler one by a single component, so each is compared with its simpler form at the settings of Section efsec:comparison: K-Means++ adds a seeding rule to K-means (A), HDBSCAN adds a density hierarchy to DBSCAN (B), GMM adds a covariance per cluster to K-means (C), and spectral clustering depends on the graph it is given (D).

3.5.1 A: Initialisation (random vs K-Means++ seeding)

On D3 ($K = 5$) and D4 ($K = 8$) each initialisation ran from one start with 30 seeds, and each run is measured by how far its inertia lies above the best of all 60 runs. On D4, K-Means++ moves almost every start close to the best: the mean gap falls from 52.9% to 3.4% and the worst from 69.4% to 12.4%, and the mean iterations fall from 15.6 to 11.1. ARI rises from 0.430 ± 0.071 to 0.508 ± 0.076 (medians 0.42 and 0.50), but only a few starts of either kind reach ARI above 0.7 (Figure 5), and K-Means++ reaches the high group in only 2 of 30 seeds against 1 of 30. It raises the chance of a good start; one start is still not enough. On D3, 29 of 30 starts with either method reach the best inertia and one lands 28% above it; only the iterations fall, from 22.7 to 15.5, and two runs agree at ARI 0.96 with random starts and 0.98 with K-Means++.

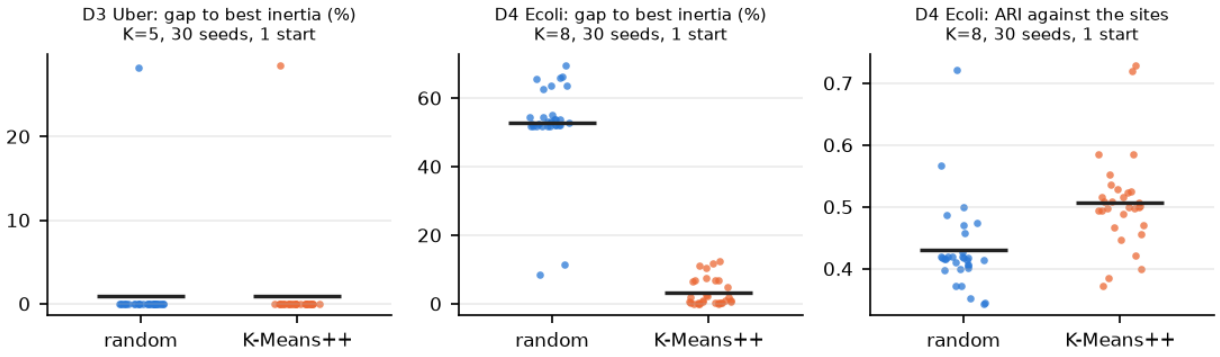


Figure 5: Ablation A, one dot per seed (30 per start method); bars mark the mean. Left and centre: inertia above the best run on D3 and D4. Right: ARI on D4. K-Means++ moves almost every D4 start near the best inertia, but only a few starts of either kind reach the high-ARI group. Cause: spreading the starting centres out makes a good local optimum more likely, not certain.

On D2 and D7 each initialisation ran from one start with 20 seeds at the chosen K. On D2 ($K = 6$) K-Means++ helps clearly: the mean gap falls from 10.9% to 4.8%, the worst from 18.8% to 11.5%, and iterations from 15.3 to 10.3. It does not remove the need for restarts. Two K-Means++

runs still agree only at ARI 0.80 on average, against 0.78 for random starts, because the belts give many local optima of similar inertia. On D7 ($K = 7$) the choice matters little: the mean gap is 2.3% with random starts and 1.3% with K-Means++, and the worst about 4% for both, though K-Means++ needs fewer iterations (23.5 against 34.9). Initialisation matters when the objective has many local optima of different quality, as on D2 and D4; where every start reaches nearly the same one, it does not.

3.5.2 B: The density hierarchy (DBSCAN vs HDBSCAN)

HDBSCAN is DBSCAN with one component added: instead of one radius for the whole dataset, it builds the hierarchy of all radii and keeps, in each region, the cluster that persists longest (Section 2). Table 6 sets the two side by side at the settings the rule chose on each dataset. The hierarchy helps where density differs between clusters. On D2 it clusters 62% of the Americas instead of 20% and calls less of the data noise overall (30.7% against 40.4%); on D1 it avoids the 72 fragments DBSCAN settles on (ARI 0.009 to 0.709); on D3 it separates LaGuardia from the city. It does not help where there are no gaps at any density: on D4 both pick the same partition, and on D5 and D7 ARI stays below 0.08. On D6 it is worse (0.431 to 0.260), because it calls 42% of the wines noise. Its cost is time: it is slower on every dataset, by a factor of about 24 on D2.

Table 6: Ablation B: DBSCAN against HDBSCAN at the settings chosen without labels. ARI on labelled data; D2 and D3 have no labels.

	DBSCAN (one radius)				HDBSCAN (hierarchy)			
Data	**Clusters**	**Noise**	**ARI**	**Time (s)**	**Clusters**	**Noise**	**ARI**	**Time (s)**
D1	72	47.1%	0.009	0.002	2	15.6%	**0.709**	0.012
D2	41	40.4%	–	0.044	28	30.7%	–	1.041
D3	2	2.1%	–	0.482	3	4.8%	–	1.271
D4	2	0.3%	0.038	0.002	2	0.3%	0.038	0.002
D5	2	22.7%	**0.028**	0.019	2	32.9%	0.023	0.384
D6	2	23.6%	**0.431**	0.001	2	42.1%	0.260	0.002
D7	2	2.6%	0.033	0.517	2	31.6%	**0.076**	2.442

3.5.3 C: A covariance per cluster (K-Means++ vs GMM)

A Gaussian mixture with equal spherical covariances and hard assignments reduces to K-means; the component GMM adds is a covariance matrix per cluster, which lets clusters differ in size, shape and orientation (Section 2). At the same K on every dataset, the covariance raises ARI on four of the five labelled datasets, most on D4 (0.508 to 0.618) and D5 (0.343 to 0.389), and slightly on D1 and D6 (0.481 to 0.501, 0.897 to 0.915); on D7 it lowers it (0.173 to 0.158). It does not repair a wrong shape: on D1 both still cut the moons. And the silhouette falls on every dataset except D1, where it is unchanged (on D4 from 0.260 to 0.157), so an analyst choosing by the silhouette would remove the component exactly where it helps most.

3.5.4 D: The affinity graph (spectral clustering on D6)

Spectral clustering partitions whatever graph it is given, so the affinity is its key component. On D6 it ran twice at $K = 3$ on the same standardised data: once with a dense RBF kernel at scikit-learn's default bandwidth ($\gamma = 1$), and once with the 10-nearest-neighbour graph used on every other dataset. ARI is 0.445 with the kernel and 0.880 with the graph (silhouette 0.249 and 0.283). The likely reason is scale: with 13 standardised features most pairs of wines are several units apart, so a kernel with $\gamma = 1$ gives them almost no affinity and the graph carries little of the class structure, while the neighbour graph adapts to the local distances. The same graph is what lets spectral clustering recover the moons of D1 exactly (Section 3.4.1).

Table 7: Ablation C: K-Means++ against GMM at the same K (the rule's K). S: silhouette; GMM's covariance type, chosen by BIC, in the last column.

	K-Means++		GMM		
Data	**S**	**ARI**	**S**	**ARI**	**Covariance**
D1 ($K = 2$)	0.495	0.481	0.495	**0.501**	tied
D2 ($K = 6$)	0.529	–	0.407	–	full
D3 ($K = 5$)	0.481	–	0.454	–	full
D4 ($K = 8$)	0.260	0.508	0.157	**0.618**	full
D5 ($K = 10$)	0.202	0.343	0.129	**0.389**	full
D6 ($K = 3$)	0.285	0.897	0.280	**0.915**	diagonal
D7 ($K = 7$)	0.257	**0.173**	0.126	0.158	full

3.6 Runtime and scalability

D7 is large enough to measure how each method grows. K-means, DBSCAN and HDBSCAN ran at fixed settings on random subsets of 1,000 to 234,292 sales (K-means from one start). Figure 6 plots the median time against size on log scales, where the slope of a line estimates the exponent b in $t \propto n^b$ (fitted from 5,000 sales up). K-means is far cheaper than either density method at every size: under 0.08 s on all 234,292 sales. Its fitted slopes (0.85 and 0.64) are below 1 only because a fixed start-up cost dominates at these sizes; each iteration is linear in n. HDBSCAN grows with slope 2.04, from 0.007 s at 1,000 sales to 222 s at 234,292. DBSCAN grows with slope 1.71, but above 50,000 sales it was not run at all, because of memory rather than time. With a fixed ε, every sale has more neighbours as the data grow: estimated from probe points, 100,000 sales would hold about 178 million neighbour pairs and the full data about 943 million, some 7.5 GB for the neighbour indices alone, which scikit-learn's DBSCAN stores in full. HDBSCAN, slower at every size, did run on all sales.

D5 was sampled down from 70,000 to 3,000 images because three methods scale badly with n in 50 dimensions. Spectral clustering needs a graph over all points and an eigendecomposition that grows faster than linearly, and a dense affinity matrix at $n = 70{,}000$ would take about 39 GB in double precision. DBSCAN and HDBSCAN rely on neighbour queries, and in 50 dimensions spatial indexes prune little, so these queries approach the $O(n^2)$ brute-force cost. On the 3,000-point sample one fit takes 0.02 s for DBSCAN, 0.34 s for K-Means++, 0.38 s for HDBSCAN, 1.4 s for spectral clustering and 34 s for GMM. GMM is by far the slowest here: with full covariances each EM step costs $O(nkd^2)$ with $d = 50$, and it runs three starts. These times exclude parameter search, which refits each method once per setting tried.

On the other datasets one fit takes at most a few seconds. On D3 (15,000 points) it takes 0.03 to 0.04 s for K-means, 0.12 s for GMM, 0.19 s for spectral clustering, 0.48 s for DBSCAN and 1.3 s for HDBSCAN; on D7 (20,000 sales) GMM takes 1.2 s and spectral clustering 0.5 s; on D4 every method takes under 0.05 s (Tables 3–5). At these sizes runtime does not decide the choice of method; at the size of the full D7, it does.

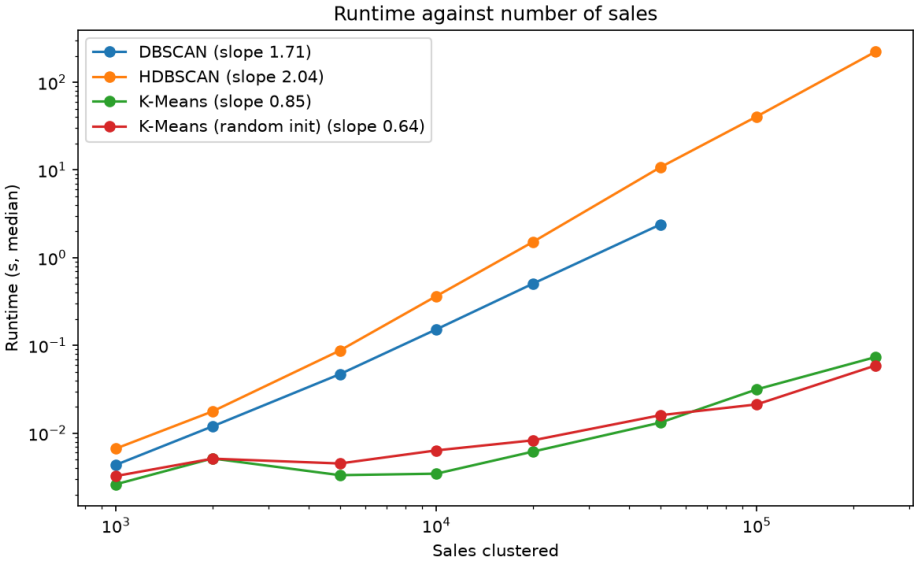


Figure 6: D7, median fit time against the number of sales, log–log, at fixed settings; the slope of each line is in the legend. DBSCAN stops at 50,000 sales: the full data would need about 7.5 GB of neighbour indices.

3.7 Success and failure cases

3.7.1 Successes

HDBSCAN on D2: belts of varying density. Shown in Figure 2. HDBSCAN keeps the sparse belts of the Americas (62% of earthquakes clustered, against 20% for DBSCAN) as long clusters that follow the plate boundary. It succeeds because its density threshold adapts to each cluster, the property D2 needs most.

Spectral clustering on D1. Spectral clustering follows the curve of each moon and matches the labels exactly (ARI 1.000; Figure 7). It succeeds because each moon is a chain of close neighbours, and the gap between the moons is wider than any link in the chain, so the neighbour graph falls apart into the two moons. K-Means++ and GMM fail on the same data because they assume convex clusters.

K-Means++ and GMM on D4: compact classes in seven dimensions. At the number of classes, GMM (ARI 0.618) and K-Means++ (0.508) match the localisation sites best of the six methods; with $K = 7$, K-Means++ reaches 0.731, the best result on D4. At that setting it keeps 96% of the largest class (cytoplasm, 143 proteins) in one cluster, 86 to 91% of the three mid-sized classes, and all 5 proteins of the outer-membrane lipoproteins together. They succeed because the classes are compact groups in seven dimensions that touch, with no density gap between them: the case centroid methods are built for, and the case where a density threshold fails (Section 3.4.3). The inner-membrane proteins (77) are the weak spot: they overlap the related inner-membrane classes, and only 52% end up in one cluster.

HDBSCAN on D7, floor area and lease only: flat designs. This feature set was chosen after seeing the main results, so we present it as an ablation finding and a sanity check, not as a tuned setting. Dropped to two features, D7 does contain dense spots: standard flat designs, built to fixed plans in particular years. HDBSCAN finds 40 clusters (25.3% noise), and 37 of them are at least 97% one flat type, 27 of them entirely so (Figure 8); in the median cluster one flat model accounts for 82% of sales. A simple grouping by flat model and lease would give similar groups,

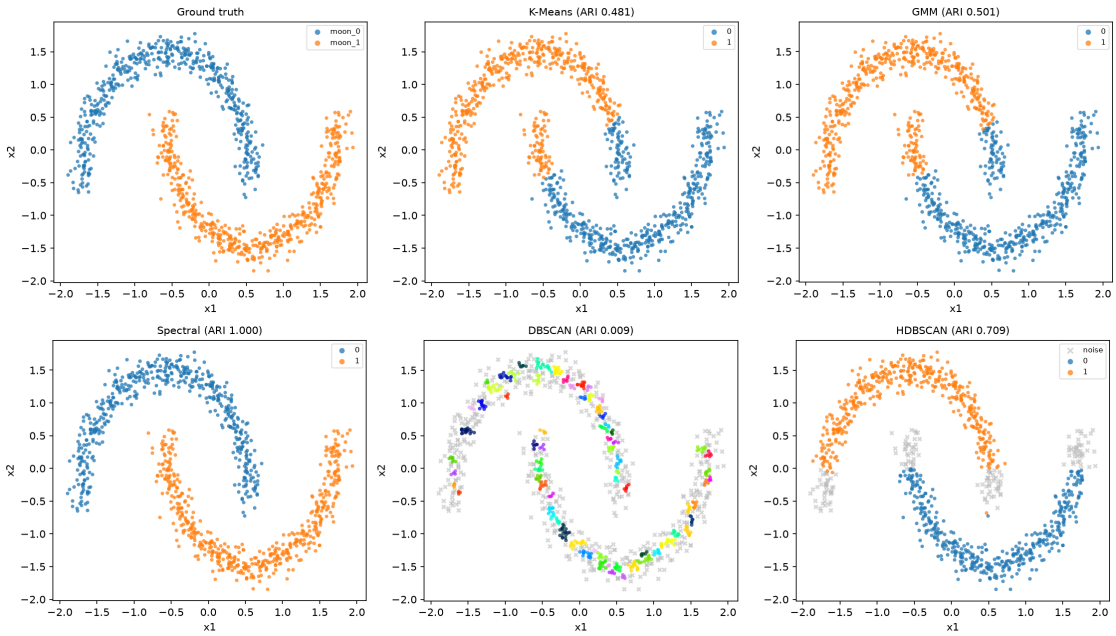


Figure 7: D1 make_moons at the settings chosen without labels. K-Means++ and GMM cut both moons with a straight boundary (ARI 0.48, 0.50); spectral clustering follows the curves (ARI 1.00). DBSCAN breaks the moons into 72 fragments (ARI 0.01) and HDBSCAN drops the sparse tips as noise (ARI 0.71), because the silhouette rewarded those settings. Grey crosses: noise.

which is the point: the method recovers a structure we can name. ARI against flat type is still only 0.081, because each type is split into several designs and the noise counts as a cluster. A low ARI here reflects a finer partition than the labels, not a failure. On shuffled columns the same setting finds 38 clusters that match nothing (ARI 0.001).

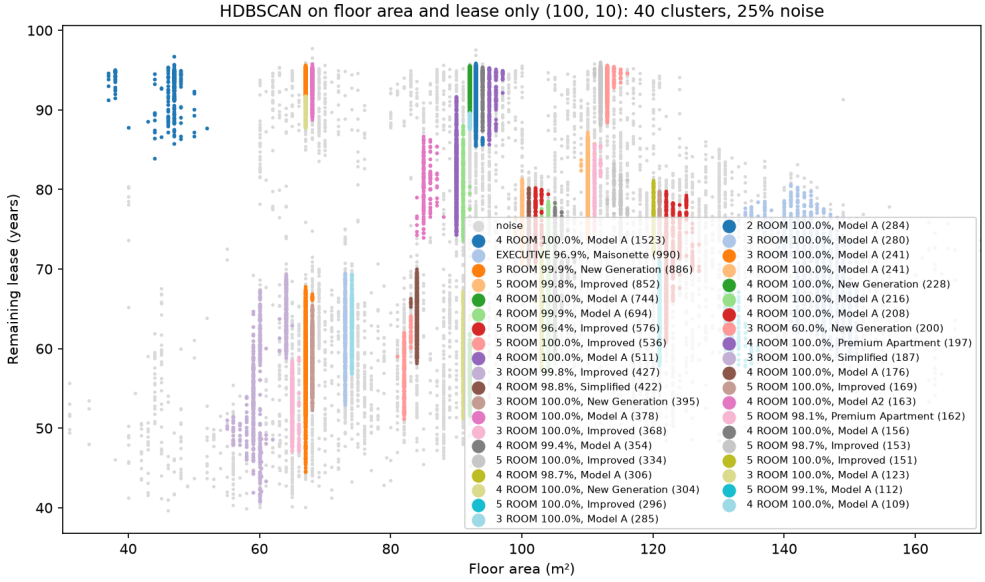


Figure 8: D7, success of HDBSCAN on floor area and remaining lease only (`min_cluster_size` 100, `min_samples` 10; grey: noise). Each cluster is one standard flat design; the legend gives its main flat type and model. Cause of success: fixed building plans create real dense spots in these two features.

3.7.2 Failures

K-Means++ on D2: unrelated belts merged. Shown in Figure 1. One cluster joins five belts whose centres are 2,932 to 11,449 km apart, from Iceland to the Mid-Indian Ridge. Cause: cluster shape. Compact regions cannot follow long belts.

DBSCAN on D1: the silhouette's choice. Shown in Figure 7. Among the settings within the noise cap, the silhouette is highest for a radius so small that the moons break into 72 fragments, with 47% of points called noise (ARI 0.009). A wider radius recovers both moons exactly, with a lower silhouette. Cause: the silhouette rewards small, tight clusters and ignores the noise, so the rule prefers fragments to the true curved shapes.

Density methods on D7: one mass, or storey bands. Shown in Figure 3. At the chosen settings DBSCAN puts 96% of sales in one cluster. On the recorded values HDBSCAN returns 37 clusters, 34 of them single storey bands. Causes: no gaps between groups, and values recorded in steps.

DBSCAN on D5. DBSCAN puts three quarters of the images, from nearly every class, into one cluster and 23% into noise (ARI 0.028; Figure 9). The cause is one global ε in 50 dimensions: an ε large enough to connect points within a class also connects the overlapping classes to each other, and a smaller one turns most points into noise. The k-based methods, which are not tied to a density threshold, reach ARI 0.34–0.40 on the same data.

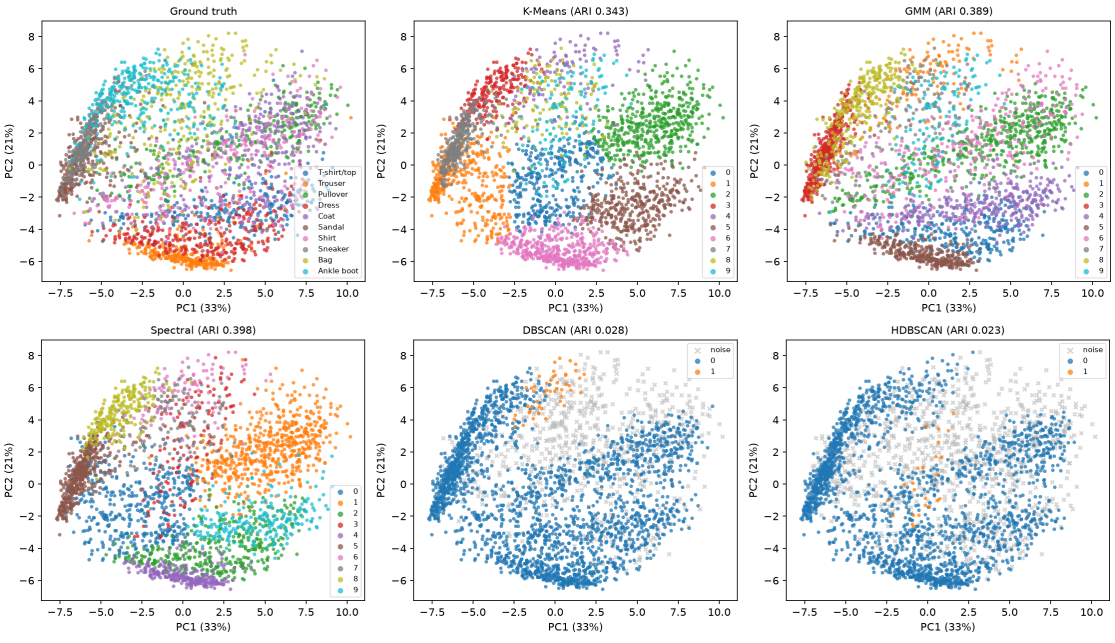


Figure 9: D5 Fashion-MNIST (3,000 images; clustered in 50 PCA dimensions, plotted in the first two). DBSCAN and HDBSCAN put most images from nearly every class into one cluster (ARI 0.03 and 0.02); spectral clustering, GMM and K-Means++ reach ARI 0.40, 0.39 and 0.34. Grey crosses: noise.

HDBSCAN on D4: rare classes called noise. Even at its best setting by ARI, a minimum cluster size of 12, HDBSCAN returns two clusters and calls 37% of proteins noise (ARI 0.339), with a silhouette (0.483) higher than that of any method that assigns every protein. The noise is not spread evenly. It takes 95% of the outer-membrane class (20 proteins), every protein of two of the three smallest classes (5 and 2), and about half of the inner-membrane and periplasm classes. Cause: a class of 2 to 20 proteins in seven dimensions is not dense enough to form a cluster of 12,

so HDBSCAN discards exactly the classes that are hardest to find, and the silhouette, which skips noise, rewards it. The setting the rule picks without labels avoids the noise only by putting 97% of proteins in one cluster (ARI 0.038).

3.8 Key factors and practical guidance

Key factors. Table 8 ranks the factors by the largest change each one caused in our experiments; the evidence is in the section named in the last column. Two patterns stand out. The first three factors decide *which family* of method can work at all, and no parameter setting overrides them. The rest change how well a suitable method does.

Table 8: Key factors, ranked by the largest effect we measured.

	Factor	Methods it hurts	Largest effect measured	Evidence
1	Cluster shape vs the method's assumption	K-means, GMM	D1: ARI 0.48 against 1.00 for spectral	§3.4.1
2	No gaps between groups	DBSCAN, HDBSCAN, spectral	D4, D7: one cluster holds 96–97%	§3.4.2, §3.4.4
3	Number of dimensions	DBSCAN, HDBSCAN	D5: no density setting reaches ARI 0.07	§3.4.3
4	Selection by the silhouette	all, most of all density	D1: DBSCAN ARI 0.009 chosen over 1.000	§3.4.1
5	Density varying between clusters	DBSCAN	D2: Americas 20% clustered against 90% elsewhere	Abl. B
6	Values recorded in steps	DBSCAN, HDBSCAN	D7: 34 of 37 clusters are storey bands	§3.4.4
7	Feature scale	K-means, GMM	D7: raw price holds 100% of distance	§3.2
8	Initialisation	K-means	D4: mean gap to best inertia 52.9%	Abl. A
9	Data size	DBSCAN, HDBSCAN	D7: DBSCAN out of memory above 50,000	§3.6

Practical guidance.

1. **Choose by the structure of the data, not by a score.** Plot the data or a projection first. If the groups are curved or long, no choice of k will help K-means or GMM; if they are compact and touching, or the data have more than a handful of dimensions, no choice of ε will help DBSCAN.
2. **Do not pick settings by the silhouette alone.** It favours few, round clusters, small fragments and settings that call hard points noise. First restrict the search to what the analysis needs (a noise cap, a minimum number of clusters), then compare within that range.
3. **Report the noise fraction and the largest cluster next to every silhouette.** A silhouette computed after discarding half the points, or with almost all of them in one cluster, says little.
4. **Compare every result against fake data.** Rerun the chosen setting on shuffled columns or uniform points. It catches results with no real structure, though not results with the wrong structure.
5. **Check for values recorded in steps.** Tied values form dense spots that density methods report as clusters; add jitter within each step and see whether the clusters remain.
6. **Standardise before K-means, and use K-means++ with several starts.** One start is not enough when the objective has many local optima.
7. **Prefer a parameter whose unit means something, and measure distance correctly.** A radius in kilometres can be checked by someone who knows the area; a k cannot. Use distance on the sphere for global coordinates; a local projection is enough within a city.

4 Conclusion

We compared six clustering methods, K-means with random and K-means++ starts, GMM, spectral clustering, DBSCAN and HDBSCAN, on seven datasets, each chosen to test one data property, with every setting chosen by one label-free rule. Which family wins depends on the data. Density

and graph methods win when groups are separated by gaps and have any shape, in few dimensions: on D1 spectral clustering recovers the moons exactly where K-means reaches ARI 0.48, and on D2 HDBSCAN follows belts that K-means merges. Centroid and mixture methods win when classes are compact and touch, or the data have many dimensions: on D4 GMM reaches ARI 0.618 and K-Means++ 0.508 while both density methods fall to 0.04, and on D5 no density setting reaches 0.07. Where the data have no gaps at all, as in the core of D3 and in D7, no family finds natural groups. Density methods return one dominant cluster, spectral clustering scores below its own fake data, and K-means draws borders where its objective puts them.

The silhouette did not tell us which case we were in. On four of the five labelled datasets the method with the top silhouette was not the one that matched the labels best; on D4 it ranks the six methods almost in reverse (Spearman $\rho = -0.82$); and on D1 it chose a DBSCAN setting that shattered the moons when another setting recovered them exactly. Rerunning each result on fake data helps where a method finds nothing real (D4, D7), but not where it finds the wrong structure (D2). Adding one component to a method, namely K-means++ seeding, HDBSCAN's density hierarchy, GMM's covariances and spectral clustering's neighbour graph, repaired the failures it targets, each at a cost, and none produced clusters in data with no gaps. The choice of method should follow from what the data look like and what the analysis needs, and every score should be checked against fake data and against the points it ignores.

Limitations

D2 and D3 have no labels, so their clusters are judged against plate-boundary geography and the map of the city only. Density parameters such as `minPts` and `min_cluster_size` are counts of points, so the clusters change with the size of the data: a random half of the 2023 earthquakes agrees with the full year at ARI 0.51 to 0.55 for the density methods, against 0.97 to 0.99 for K-Means++. D7 has no location beyond the town, so its clusters describe flats rather than places, and its one label, flat type, is nearly a function of floor area, so a low ARI on D7 says little about the quality of a clustering.

On the labelled datasets, K-means, GMM and spectral clustering were given the number of classes, which an analyst would not know, and the best-ARI rows use the labels outright; both overstate what can be reached without labels. D4 and D6 are small (336 and 178 points), so a single protein moves a rare class's share by up to 50%. The datasets keep their owners' parameter grids, so the rule is shared but the search space is not. On D1 and D6 the fake-data setting is chosen again on the fake data, and on D5 the fake-data check was not run, because its data could not be rebuilt identically outside the owner's environment.

Our evidence on dimensionality comes from two datasets, D4 with 7 features and D5 reduced to 50 principal components from 784 pixels, which keep 86.5% of the variance. We did not test the methods on the full 784 dimensions or beyond, so the high-dimensional results describe data that have already been reduced.

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